
Partial Differential Equations

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Part of a series of notes available at
kalaapks.github.io/notes.html

See kalaapks.github.io/notesDE.html for the notes on DEs.

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FIRST ORDER PDEs

1

1.1 Method of characteristics

Since we will be solving for surfaces u in PDEs, we have to find a way to simplify the geometric structure of u or somehow make the problem itself abstract so that we can eliminate the need for geometry altogether. Turning the problem abstract comes with its own complications and before seeing them we have first have to explore the limitations of the geometric approach. Luckily, everything we do in the first chapter will involve geometry and that is because first order PDEs do not pose threat as great as the second order ones (with which we deal in Chapter 2).

For the linear PDEs of first order, we plan to decompose the domain where the PDE is defined into family of curves and then solve for u on each one of these "strings". We call these curves the 'Characteristic curves' and the process as 'Method of characteristics'.

Most of the processes we actually see in nature will need 2^{nd} order PDEs but still understanding the 'Method of characteristics' is very crucial for one reason. You will notice, once we have done the theory, that this method exploits a certain feature that only the surfaces describable by 1^{st} order PDEs have. Given the initial conditions they simply are transported forward in time. They do not behave as erratic as the higher order PDEs.

1.1.1 Constant coefficient Transport equation

We will start with a simple PDE. Why this is a simple form for a PDE will get clearer down the line as we see more complicated models. For now just notice this is linear and involves only first order derivatives.

$$\langle b, \nabla u \rangle + u_t = 0$$

$$\text{or, } \sum_{j=1}^n b_j \frac{\partial}{\partial x_j} (u) + u_t = 0$$

Alternate representation: For $b \in \mathbb{R}^n$, $(b, 1) \in \mathbb{R}^{n+1}$, then,

$$\langle (b, 1), (\nabla u, u_t) \rangle = 0 \quad (1.1)$$

Remark 1.1. $\nabla u = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right)$ will denote the 'space-derivatives' and hence $(\nabla u, u_t) = \left(\frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_n}, \frac{\partial u}{\partial t} \right)$ is simply the space-time gradient of u .

Naturally, equation $\langle (b, 1), (\nabla u, u_t) \rangle = 0$ from 1.1 means space-time gradient of u along $(b, 1)$ vector is 0. This is the essence of "homogeneous" PDE. Now, for a point (x, t) in $\mathbb{R}^n \times \mathbb{R}$ and for some $s \in \mathbb{R}$: $(x, t) + s(b, 1) = (x + sb, t + s)$ is a straight line in $\mathbb{R}^n \times \mathbb{R}$ through point (x, t) and in the direction of $(b, 1)$.

Notice what happens along these lines. Denote one of them as $z(s) = u(x + sb, t + s)$. Then,

$$z_t(s) = \langle \nabla u(x + sb, t + s), b \rangle + u_t(x + sb, t + s) \quad (1.2)$$

However, 1.1 gives us that the above quantity is $0 \Rightarrow z_t(s) = 0 \Rightarrow z$ is a constant function. **The solution surface $u(x, t)$ in $\mathbb{R} \times (\mathbb{R}^n \times \mathbb{R})$ is constant along these s -lines in domain $\mathbb{R}^n \times \mathbb{R}$ and these s -lines are called Characteristic lines.** Recall, we mentioned the current method will exploit a certain feature found only in the surfaces described by 1st order PDEs. This is that feature in this case (we say 'in this case' because 1.1 is not the most general form a 1st order PDE).

Geometry of characteristic lines for $\langle (b, 1), (\nabla u, u_t) \rangle = 0$: They have the form:

$$\{(x, t) + s(b, 1) : s \in [0, \infty)\} \text{ for } x \in \mathbb{R}^n$$

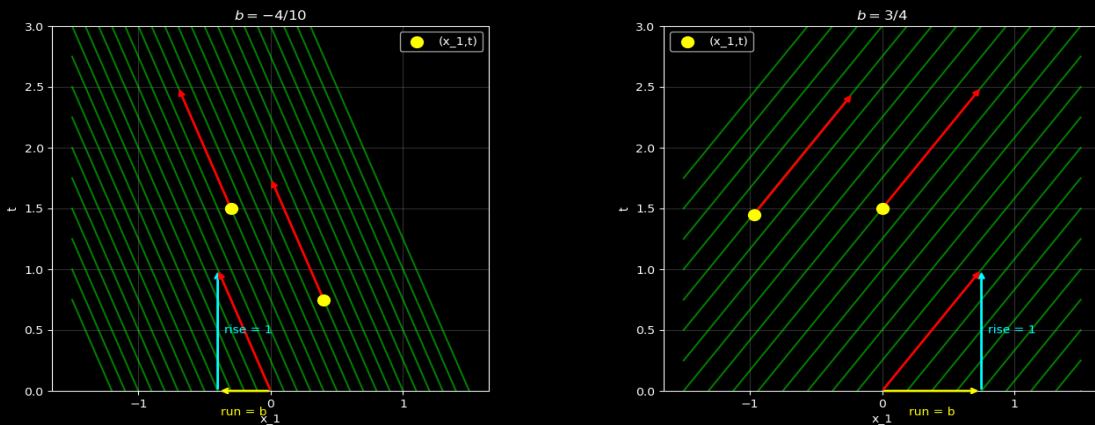


Figure 1.1: Characteristic lines in 2-D

We start with $n = 1$, that is, only one space variable and the other is time variable. Then, b is a scalar and hence can be either negative or non-negative. We can plot them separately and exhaust all possibilities. All these characteristic lines tells us is if we start at any point (x_1, t) , we must move along $(b, 1)$.

Similarly, for $n = 2$, that is, two spatial variables and one time variable now. Characteristic lines still says the same story - if we start at any point (x_1, x_2, t) , we must move along $(b, 1)$.

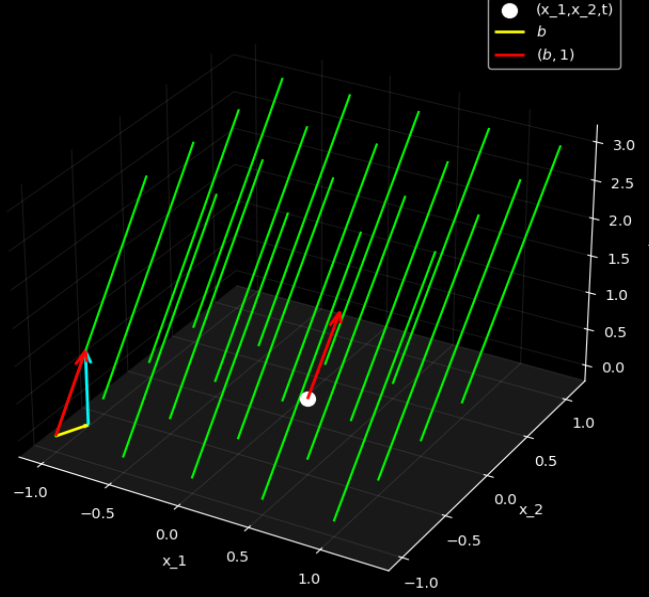


Figure 1.2: Characteristic lines in 3-D

Analogous idea follows for higher dimensions.

[Question] So far, we have gotten hold of the family of lines on which u is constant. Aren't these the contour lines then?

Yes, almost. The only difference is we simply know u is constant on these curves, we do not know what value u takes on them yet. For that, we need to get the values at atleast one point on every characteristic curve. That brings us to the Initial Value Problems. We will assume that the value u takes at $t = 0$ on every single one of the Characteristic curves is known.

Consider the hypersurface $\Sigma := \{t = 0\}$ in $\mathbb{R}^N \times \mathbb{R}$ and let $\bar{u} \in \mathcal{C}^1(\Sigma; \mathbb{R})$. We look for solution to the following problem:

$$\begin{cases} u_t + \langle b, \nabla u \rangle = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ u \in \mathcal{C}^1(\mathbb{R}^n \times (0, \infty)) \cap \mathcal{C}^0(\mathbb{R}^n \times [0, \infty)) \\ u = \bar{u} \text{ on } \Sigma \text{ for } \bar{u} \in \mathcal{C}^1(\Sigma; \mathbb{R}) \end{cases} \quad (1.3)$$

With the initial data for u given on $\Sigma = \{t = 0\}$, we call this the Initial Value Problem (IVP). Notice that an entire dimension worth of data is not given and that is our job to find. That gives us all we need though.

For any $(x, t) \in \mathbb{R}^n \times (0, \infty)$, let $\pi(x, t)$ be the point on Σ such the Characteristic line originating from $\pi(x, t)$ goes through (x, t) . **They are simply the projections along the Characteristic curves onto Σ .** We know $u(x, t)$ has the same value as $u(\pi(x, t))$. We simply need to figure out the form of $\pi(x, t)$ and we will be done. We know $\exists s \in [0, \infty)$ such that

$$(x, t) = (\pi(x, t), 0) + s(b, 1) = (\pi(x, t) + sb, s)$$

On comparing coordinate wise, we get $t = s$ and $\pi(x, t) = x - sb = x - tb$. Then, we get

$$u(x, t) = u(\pi(x, t)) \equiv \bar{u}(\pi(x, t)) = \bar{u}(x - tb)$$

We can move to $\bar{u}$ because $\pi(x, t)$ is a point in $\Sigma = \{t = 0\}$. That was the point afterall, to come down to the space where value is known. The solution of IVP 1.3 is

$$u(t, x) = \bar{u}(x - tb)$$

This reads value of u at $(x, t) \in \mathbb{R}^n \times [0, \infty)$ is equal to value of $\bar{u}$ at $\Pi(t, x) \in \{t = 0\}$ hyperspace.

1.1.2 Non-constant coefficient Transport equation

Like before we will start with the homogeneous type to make sense of the geometry involved.

Notation : $x \in \mathbb{R}^N$ and $x = (x_1, \dots, x_N)$. We will no longer distinguish between space & time variables. Untill now it was : $N = 1 + n$ and $x_1 = t$. This simply ensures that 't' no longer gets a special treatment and u_t also has a coefficient attached.

$$Lu = \sum_{i=1}^N b_i(x) \frac{\partial u}{\partial x_i} = \langle b(x), \nabla u(x) \rangle$$

$$Lu = 0 \quad \text{in } \Omega \text{ open } \subseteq \mathbb{R}^N, \quad b_i \in C^1(\Omega) \quad \forall i \in \{1, \dots, N\}$$

This is a bit more general treatment of an homogeneous 1st order linear PDE. What we did in 1 & 2 were necessary for building intuition though. This general approach comes with minor problems and questions.

1stly, $b_i(x)$ are no longer necessarily constants. Will the characteristics still be straight lines? Afterall, constant value of $b_i(x)$ gave us the constant slope.

2ndly, we had Ω as $(0, \infty) \times \mathbb{R}^n$ previously. Now, Ω is any open subset of $\mathbb{R}^N$.

3rdly, what will Σ be now? It was $\{t = 0\}$ earlier.

We can for example simply consider $\Sigma = \partial\Omega$; that is, what happened earlier because when $\Omega = \{t > 0\}$, we have $\Sigma = \partial\Omega = \{t = 0\}$. However, we have to be careful when we relax the conditions on Ω so much. Suppose Ω has the structure $\mathbb{R} \times \mathbb{R}^n$ instead of $(0, \infty) \times \mathbb{R}^n$. In this case $\partial\Omega = \emptyset$!! The choice for such Ω is not merely for a counter example either. The only reason we looked at upper half-space was because generally we are interested in future activity. This doesn't mean we shouldn't have mathematical theory for behaviour in past.

Trace back a few steps and recall the sole purpose of Σ was to be of dimension $N - 1$ so that we can get transverse curves through Σ in dimension N and call them characteristic curves.

Let $\Sigma \subseteq \Omega \subseteq \mathbb{R}^N$ be a C^1 -hypersurface without boundary and $\bar{u} \in C^1(\Sigma; \mathbb{R})$.

$$\begin{cases} Lu = 0 & \text{in } \Omega \\ u \in C^1(\Omega) \\ u = \bar{u} & \text{on } \Sigma \end{cases} \quad (1.4)$$

Earlier the hypersurface $\Sigma = \{t = 0\}$ was a hyperplane and hence $(1, b)$ was always transverse for any & every b due to 1 in the first coordinate. How do we involve the notion of transversity now that Σ can possibly be any hypersurface? Naturally, as long as $b(x)$ is not tangential to Σ , we can use the previous process with appropriate modifications.

Definition 1.2. $Char_x(L)$: Let $x \in \Omega \subseteq \mathbb{R}^N$. Then,

$$\text{char}_x(L) = \{\xi \in \mathbb{R}^N \setminus \{0\} : \langle \xi, b(x) \rangle = 0\}$$

Geometry structure : At the fixed point $x \in \Omega$, $Char_x(L)$ is the collection of all perpendicular vectors to the vector $b(x)$. We can also call it the tangent space at $b(x)$ denoted as $T_{b(x)}\mathbb{R}^N$. Recall, we were searching for directions in which the 'characteristic curves' could flow such that they are not along $b(x)$. Now we have a good candidate; any direction along $Char_x(L)$ is nice.

Definition 1.3. Characteristic at a point: Let Σ be a hypersurface of class $\mathcal{C}^1$. We say Σ is characteristic at $\tilde{x} \in \Sigma$ if $\mathbf{n}_\Sigma(\tilde{x}) \in \text{char}_x(L)$. If this is the case at every x , we say Σ is characteristic for L .

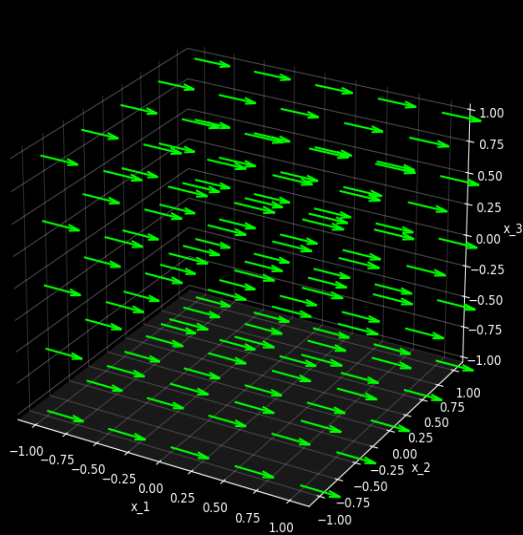
Here the structure is obvious. $\mathbf{n}_\Sigma(\tilde{x}) \in \text{char}_x(L)$ means $\mathbf{n}_\Sigma(\tilde{x}) \perp b(\tilde{x})$, that is, $b(\tilde{x}) \in T_{\tilde{x}}\Sigma$. Naturally, since we want 'transverse' we would want to avoid this alignment with the tangent space at every single point while trying to construct our required Σ .

Definition 1.4. Non-characteristic hypersurface : Let Σ be a hypersurface of class $\mathcal{C}^1$. We say Σ is non-characteristic if $\forall x \in \Sigma, b(x) \notin T_x\Sigma$.

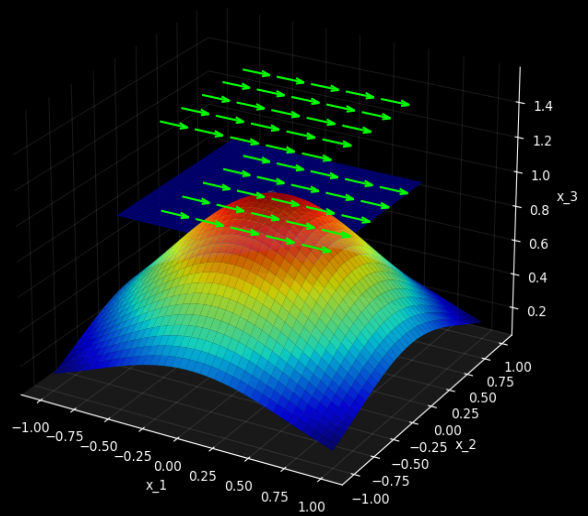
Let us see what these actually are with examples. We start with a relatively simple 3 dimensional vector field $b(x) = (1, 0, 0)$ and a piece of a surface,

$$\Sigma := \{(x_1, x_2, e^{x_1^2 + x_2^2}) : (x_1, x_2) \in (0, 1)^2\}$$

The $Char_x(L)$ is $\{x_1 = 0\}$. Naturally, it is same at every point since b is constant. As we can see in figure 1.3b, $b(0, 0, 1)$ is in the tangent plane $T_{(0,0,1)}\Sigma$. Unfortunately, because of this single point $(0, 0, 1)$, the entire surface Σ becomes non-characteristic to the PDE corresponding to $b = (1, 0, 0)$.



(a) $(1, 0, 0)$ is rescaled to $(0.25, 0, 0)$

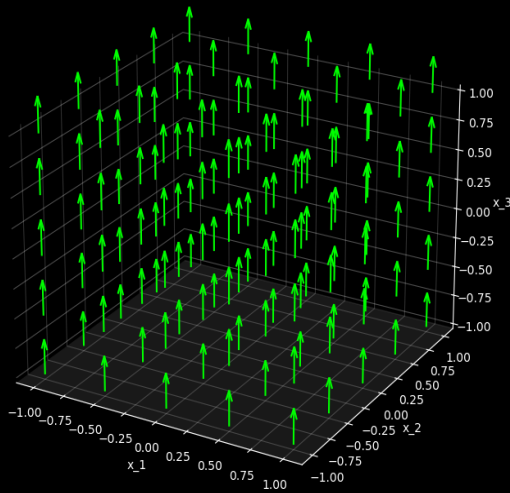


(b) Imagine a plane at $(0, 0, 1)$

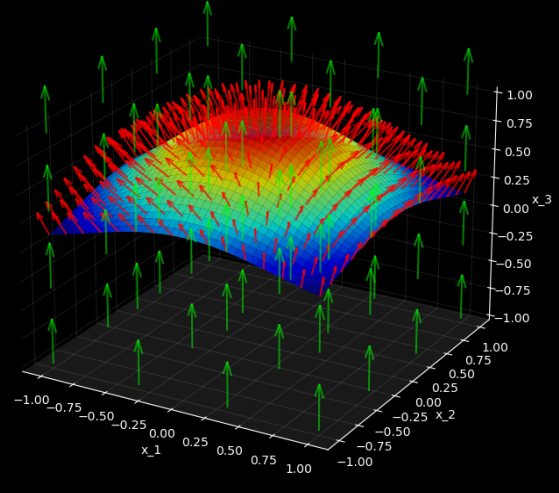
Figure 1.3: Σ is not a Characteristic hypersurface for this b

The same surface can become characteristic for a different b though. Imagining the tangent space at a point x and ensuring the green vector at that point ($b(x)$) is not in it is one of the process. The other is to look at the normal vectors (in red) on Σ and ensuring b (green vectors) are not perpendicular to them. We are going to see this method now, just so we get to explore all the possibilities. Ofcourse, for mental picture note that, when we are looking in terms of normal vectors, alignment is allowed.

Consider $b = (0, 0, 1)$ for next example. The $Char_x(L)$ is again fixed, it is the plane $\{x_3 = 0\}$. Clearly, $n_{\Sigma}(x)$ does not belong to $Char_x(L)$ for any x . Then, Σ will be characteristic here.



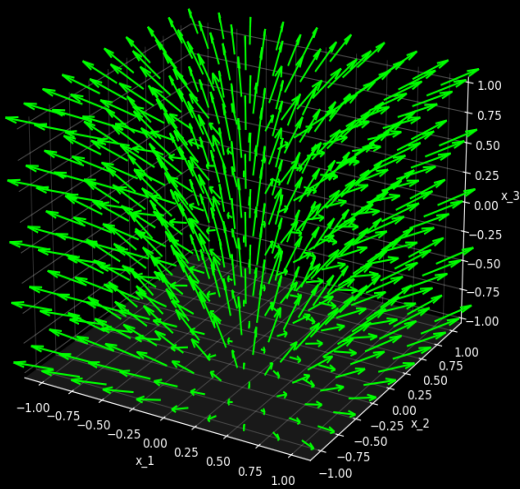
(a) $b = (0, 0, 1)$ is rescaled to $(0, 0, 0.25)$



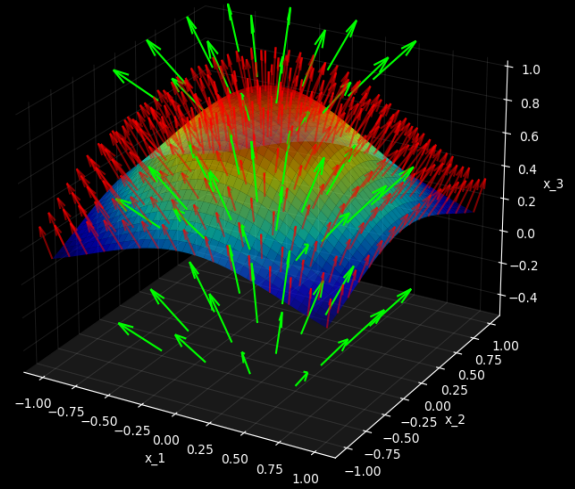
(b) The red vectors are normals ($\mathbf{n}$) to the surface

Figure 1.4: Σ is a Characteristic hypersurface for this constant b

These are all simple b -vector field. We can now consider a non-constant one too. Let $b(x_1, x_2, x_3) = (x_1, x_2, 0.5)$. This is outward going vector field with a slight nudge in the 3rd direction (x_3 here).



(a) $b(x_1, x_2, x_3) = (x_1, x_2, 0.5)$



(b) The red vectors are normals ($\mathbf{n}$) to the surface

Figure 1.5: Σ is a Characteristic hypersurface for this b

The $Char_x(L)$ here is not constant obviously.

$$Char_x(L) = sp_{\mathbb{R}} \left\{ \left(0, -\frac{1}{x_2}, 2 \right), \left(-\frac{1}{x_1}, 0, 2 \right) \right\}$$

Analytically verifying if $\mathbf{n}$ is in $Char(L)$ will be needed for such general N -dimensional $\mathbf{b}$ ($N \geq 3$).

[Question] Why do we need the surface to be non-characteristic (transvere)?

Consider $N = 2$ and the constant vector field $b(x) = (2, 0)$. Then, $\forall$ points $x \in \{x_2 = 0\} = \Sigma$ we have $b(x) \in T_x \Sigma = \Sigma$. The problem with such hypersurfaces is:

$$Lu = 2u_{x_1} + 0u_{x_2} = 0 \Rightarrow \frac{\partial u}{\partial x_1} = 0$$

Since $u = \bar{u}$ on Σ , above conditions means $\frac{\partial \bar{u}}{\partial x_1} = 0$ on Σ !! But, there should be no restriction on initial condition, they can be anything as long as $\bar{u} \in \mathcal{C}^1(\Sigma; \mathbb{R})$.

Remark 1.5. *If the hypersurface Σ is characteristic, then we end up putting restrictions on the tangential derivatives of the initial condition $\bar{u}$ which is unreasonable.*

1.1.3 The Quasi-linear variant

1.2 Non-linear first order PDEs

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SECOND ORDER PDEs

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2.3.2 Analysis of wave in n -dimensions

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FINITE ELEMENT METHOD

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